Evolution Mechanisms and Relational Role Determination — The Connecting Pair Where Ongoing Function Change Makes Level Redetermination a Governance Obligation (B1.12 + B1.17)

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 13, 2026

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026).

Abstract

The Coordination Knowledge Substrate (CKS) pattern commits to two architectural properties whose relationship is not immediately obvious: the ongoing evolution of governed AI entities through three mechanisms in productive tension (B1.12), and the determination of an entity's structural level by its current function rather than by any intrinsic fixed type (B1.17). This note formalizes the connection between them. B1.12 is the mechanism that changes entity functions over time; B1.17 is the principle that level assignment must track function. The connection is not automatic — evolution per B1.12 does not directly produce level redetermination. The connection is mediated: B2.87 evolution-triggered role change is the governance mechanism that connects significant evolution events to level-determination review. Without B2.87 as operational bridge, B1.12 evolution proceeds while B1.17 level assignments stagnate, producing the Fixed-Role Entity sub-form of Intrinsic Type Assignment (B3.18), the principal failure mode. The note develops the pair identification, the connecting mechanism through B2.87, the distinct contribution of each of the three evolution mechanisms to role-review triggering, the mutual dependency between the two commitments, co-presence requirements, operational integration, detection indicators, and the failure mode that results from the absence of the connection.

---

1. Pair Identification

Pair: B1.12 (three evolution mechanisms in productive tension) + B1.17 (relational roles, level membership functionally determined).

Relationship type: Connecting pair, mediated by B2.87 (evolution-triggered role changes).

Source sections: Paper 2 §7 (B1.12), §5.2 (B1.17), §8.2 (B2.87 operational context).

Pair position in the series: B4.23, the twenty-third Phase B4 note and twenty-second composition pair. B4.22 formalized the B1.12 internal productive-tension pair (instinct evolution and DNA evolution). B4.23 extends outward from evolution mechanics to consider how evolution interacts with the architectural principle governing structural level membership.

B1.12 states that a governed AI entity evolves continuously through three mechanisms — mutation (B1.13, instinct-layer upgrades), directed selection (B1.14, human-governed DNA modifications), and action-feedback (B1.15, lived operational experience closing back into DNA refinement) — and that these mechanisms operate in productive tension rather than through any single dominant channel. Evolution is not an episodic event in the architecture; it is the normal condition of governed entities over time.

B1.17 states that an entity's membership in a given structural level — cell, aspect, or Self — is determined by what the entity *does*, not by what it *is* in any fixed categorical sense. Level assignment is made per B2.85 (purpose-defined level determination) based on functional profile at the time of assessment. Because level is relational and function-derived rather than intrinsic and type-fixed, it is in principle revisable whenever functional profile changes sufficiently to warrant it.

The connection between the two commitments is: B1.12 produces functional change; B1.17 requires that level assignment reflect current function; therefore B1.12 evolution events can generate the conditions under which B1.17 mandates level review. B2.87 evolution-triggered role change names the governance activity that operationalizes this requirement.

---

2. Enabling Direction — How Each Commitment Enables the Other

B1.12 provides the function-changing mechanism that B1.17 must track. Without ongoing evolution per B1.12, level determination per B1.17 and B2.85 would be a one-time event at birth with no subsequent occasion for review. An entity born as a cell would remain a cell not because its function justified that assignment indefinitely, but because nothing had changed its function. B1.17's relational principle — that level is function-determined — would be architecturally true but operationally dormant. B1.12 makes B1.17's revisability principle architecturally live by continuously generating the functional changes that may call for level reassessment.

Each of the three mechanisms contributes differently to this. Mutation per B1.13 arrives from upstream — LLM provider upgrades and substrate-platform infrastructure changes — and is undirected from the entity's perspective. The functional changes mutation produces are not planned; they arrive as capability jumps that may expand or contract what the entity effectively does at the instinct layer. Directed selection per B1.14 is the most deliberate mechanism: human governance adds, removes, or restructures DNA rules, potentially expanding operational territory significantly, redirecting the entity to different functions, or adding coordination behaviors the entity did not previously carry. Action-feedback per B1.15 is empirical: accumulated operational evidence may show systematic patterns in what the entity actually does day-to-day, which may differ from its formally assigned scope in ways that warrant structural reconsideration.

B1.17 provides the normative basis that makes evolution review architecturally obligated rather than optional. B1.12 by itself specifies that evolution occurs and is governed; it does not specify that evolution has implications for level assignment. B1.17 provides that implication: because level should reflect current function, and because current function changes through B1.12 mechanisms, level assignments become increasingly inaccurate as evolution proceeds without review. The obligation to review is not merely practical — it is what B1.17's functional-determination principle entails. If level were intrinsically fixed, evolution would have no bearing on level assignment; because level is relational and functionally derived, evolution that changes function is directly relevant to level accuracy.

---

3. B2.87 as Operational Intersection

B2.87 evolution-triggered role change is where B1.12 and B1.17 meet in operational practice. It is not a commitment asserted by either parent note independently; it is the specifically architectural mechanism that the *conjunction* of B1.12 and B1.17 requires.

The mechanism operates in three phases. First, a significant evolution event occurs through one of the B1.12 mechanisms — a mutation that substantially shifts behavioral scope, a directed selection event that expands operational territory, or action-feedback evidence showing systematic scope divergence. Second, the significance of the event is assessed against the question B1.17 poses: has the entity's functional profile changed enough that its current level assignment per B2.85 no longer accurately describes it? Third, if governance determines that the entity's current function better fits a different level, level redetermination is authorized through a directed selection event per B1.14, a new level determination record is created per B2.85, and subsequent governance applies commitments appropriate to the new level per B1.20.

B2.87 does not trigger level redetermination automatically; it triggers governance review. The distinction matters. Automatic level shifting on every evolution event would create architectural instability — entities oscillating between level assignments as functions shift incrementally. Governance-triggered review requires human judgment about whether the functional change is substantial enough, whether the new level assignment is warranted, and whether the organizational consequences of level change are appropriate at this time. B1.17's relational principle is satisfied by accurate determination when performed, not by instantaneous automatic adjustment.

The threshold question — how much functional change triggers a B2.87 review — is a governance decision rather than an architectural specification. The architecture commits that significant changes warrant review; it does not specify a formula for significance. Governance holds this judgment at the same authority scope that holds DNA evolution decisions per B1.14.

---

4. Three Mechanism Contributions to Role Review

The three evolution mechanisms contribute differently to the conditions under which B2.87 review becomes warranted.

Mutation per B1.13 primarily affects behavioral characteristics at the instinct layer. Because mutation arrives from upstream and is undirected, the functional changes it produces may be difficult to anticipate. An LLM upgrade may sharpen the entity's reasoning in ways that cause it to operate at a scope significantly beyond its previous level assignment — effective coordination behaviors emerging where none existed. B2.87 review triggered by mutation tends to be reactive: governance observes behavioral change and assesses whether the functional profile has shifted sufficiently. The detection signal is typically provided by action-feedback evidence (B1.15 operational patterns) noticing that post-mutation behavior diverges from pre-mutation norms.

Directed selection per B1.14 is the most likely trigger for role-change assessment, and the most tractable to catch at the right moment. Because directed selection involves deliberate DNA modifications under human governance, the events are documented, reviewed, and consequential by design. When a directed selection event adds coordination functions to a cell — functions that, considered together, now describe what an aspect does rather than what a cell does — the appropriate governance question is whether the entity should be redetermined at the aspect level. B2.72 directed selection event review is the natural checkpoint: any directed selection event that materially expands operational territory or adds functions characteristic of a higher level should include a B2.87 role-change assessment as a component. Because directed selection is where governance most intentionally changes what an entity does, it is where the connection between B1.12 and B1.17 is most explicitly exercisable.

Action-feedback per B1.15 contributes a distinct kind of evidence: systematic operational patterns may surface that an entity is regularly doing things outside its determined scope. The canonical case is an entity assigned at the cell level whose accumulated action-layer records show persistent coordination behavior — patterns that are characteristic of aspect-level function rather than cell-level function. This divergence does not require that any single evolution event produced the scope shift; it may have accumulated gradually through multiple small directed selection events, each of which seemed incremental in isolation. Action-feedback evidence functions as an audit mechanism on level accuracy: it can surface discrepancies between formal level assignment and functional reality that directed selection review alone might not catch. B2.87 review triggered by action-feedback evidence typically leads to a broader assessment of how the entity's function has evolved since its last level determination.

---

5. Mutual Dependency

B1.12 depends on B1.17 for level coherence through evolution. Without B1.17's relational determination principle, evolution per B1.12 produces functional change without any architectural obligation to review level assignment. Entities accumulate function changes through mutation, directed selection, and action-feedback without triggering the governance assessment that would determine whether their level assignment remains accurate. The absence of B1.17 means that level assignment is implicitly treated as intrinsic — fixed at birth, not revisable through evolution — even if no explicit commitment to that position has been made. B1.17's relational principle is what provides the normative basis for B2.87 review; without B1.17, B2.87 has no architectural motivation.

B1.17 depends on B1.12 for ongoing governance relevance. An architecture in which entities do not evolve would make level determination a one-time activity. B1.17's relational principle would still be logically true — level is function-determined — but operationally irrelevant because functions never change. B1.12's continuous evolution is what makes B1.17's principle a live architectural obligation rather than a dormant theoretical commitment. The architecture does not just say that level is revisable in principle; B1.12 ensures there are regular occasions on which revision must be considered.

The mutual dependency has a specific consequence for governance process design. Because B1.12 evolution is continuous and B1.17 accuracy is therefore continuously at risk of degradation, governance processes must include evolution-awareness as a standing component of level review. It is not sufficient to determine level accurately at birth and never revisit it; the B1.12 / B1.17 pair together require that significant evolution events be assessed against level accuracy on a recurring basis.

---

6. Co-Presence Requirement

B1.12 without B1.17 awareness (evolution without role review). When B1.12 evolution proceeds without the B1.17 principle triggering level review, entities accumulate functional changes without structural adjustment. The Fixed-Role Entity sub-form of Intrinsic Type Assignment (B3.18) develops — the entity is treated as having the level it was assigned at birth, regardless of how its function has evolved. Governance applies level-appropriate Paper 1 commitments at scope that no longer matches the entity's actual operational reach. A cell-level entity operating with aspect-level scope is governed as though its reach were bounded to cell-level concerns; the governance machinery is applied at the wrong scope, and the entity's actual operational impact is not subject to the authority and accountability that its effective function warrants.

B1.17 without B1.12 evolution (relational determination without evolution consideration). When level assignments are correctly determined at birth per B2.85 but governance processes do not connect evolution events to level review, accuracy degrades silently. The determination was correct at origination; it becomes less accurate with each evolution event that changes functional profile without triggering review. This is not the same failure mode as B1.12 without B1.17 — the entity's level is not locked by intrinsic assumption, but by process gap. The architectural commitment to relational determination is present; the operational connection from evolution to review is absent.

Full co-presence. When B1.12 and B1.17 operate together with B2.87 as the connecting mechanism, significant evolution events trigger governance review of level assignments; level assignments remain current through ongoing evolution; governance applies commitments at the scope appropriate to the entity's actual function at every stage of its lifecycle. The architecture is coherent through time, not only at birth.

---

7. Operational Integration

The operational sequence for the B1.12 / B1.17 connection through B2.87 proceeds as follows.

An entity is born with a level determination per B2.85, based on functional profile at origination. A determination record is created documenting the assessed function and the level assigned.

Evolution events occur through B1.12 mechanisms over the entity's operational life. Mutation per B1.13 arrives from upstream and is integrated through governance responses appropriate to the magnitude of functional change. Directed selection events per B1.14 are authorized through the standard authority architecture and reviewed per B2.72 directed selection event review. Action-feedback per B1.15 is accumulated through the action layer and reviewed through the human-mediated approval process that governs action-feedback-driven substrate changes.

At directed selection events that materially expand operational territory, a B2.87 role-change assessment is included as a component of event review. Governance reviews current function against B2.85 determination criteria. If the current functional profile better fits a different level, level redetermination is authorized through a directed selection event; a new determination record is created per B2.85; and governance applies level-appropriate commitments per B1.20 at the new level's scope going forward.

Action-feedback review per B1.15 may additionally surface evidence of systematic functional scope divergence. Where such evidence indicates a discrepancy between formal level assignment and operational function, governance initiates a B2.87 review even absent a specific directed selection trigger.

The prior level determination record is not deleted; it forms part of the entity's lineage and audit history, consistent with the architecture's commitment to retraceable provenance across all governed changes.

---

8. Detection

Three indicators allow governance to assess whether the B1.12 / B1.17 connection is functioning correctly.

B2.87 evolution-triggered role changes — governance process currency. Is there a documented governance process for reviewing level assignments when significant evolution events occur? Does that process apply specifically to directed selection events per B1.14 that expand operational territory? Is B2.87 review a named component of B2.72 directed selection event review? The absence of a named B2.87 process is the primary structural indicator that the pair connection is not operationalized.

B2.88 relational roles verification — determination record currency. Are level determination records current? Do they reflect the entity's functional profile as of its most recent significant evolution event? Stale records — determinations made at birth with no subsequent update through an entity's evolution — are the operational signature of the Fixed-Role Entity failure mode. Relational roles verification audits whether determination records have been reviewed in light of evolution history.

Pair completeness — directed selection event review scope. Do directed selection event reviews per B2.72 include explicit role-change assessment as a standing component? An event review process that evaluates the DNA modification's content and authority without assessing whether the modification changes the entity's functional level is structurally incomplete relative to the B1.12 / B1.17 pair. This is the most tractable detection point, because directed selection events are documented and reviewable.

---

9. Failure Mode

Intrinsic Type Assignment (B3.18), Fixed-Role Entity sub-form.

The primary failure mode when B1.17 does not respond to B1.12 evolution is the emergence of Fixed-Role Entities — entities whose level assignments are treated as fixed at birth regardless of functional evolution. The failure mode is named as a sub-form of Intrinsic Type Assignment (B3.18) to indicate that the failure is not merely procedural but architectural: the system implicitly commits to intrinsic type by treating level as non-revisable in practice, even if B1.17's relational principle is acknowledged in theory.

The Fixed-Role Entity sub-form is particularly dangerous when B1.14 directed selection significantly expands an entity's scope without triggering role review. The danger is not that the expansion is unintended — directed selection is deliberate and authorized. The danger is that governance has knowingly extended what the entity does into new territory while continuing to apply governance at the entity's original, narrower level. The gap between scope and governance grows with each such expansion. An entity accumulating coordination functions through successive directed selection events while remaining governed as a cell is, in effect, operating as an aspect under cell-level governance — a mismatch between operational reach and governance architecture that compounds over time.

The failure mode is recoverable. A B2.87 governance review, initiated at any point, can assess current function against B2.85 determination criteria and authorize level redetermination if warranted. Prior level determination records remain in the lineage; the redetermination does not erase the history but extends it. The cost of recovery scales with how long the gap between assigned level and actual function has been allowed to accumulate — longer gaps produce more embedded mismatches between scope and governance that must be addressed upon redetermination. Early B2.87 review, triggered at directed selection events that materially change operational territory, is structurally cheaper than deferred redetermination triggered after extensive functional drift.

The general lesson the pair teaches is that level determination is not a one-time governance event but an ongoing governance obligation. The architecture makes level revisable because functions evolve; making revisability real requires that governance processes connect evolution events to determination review through B2.87. The B1.12 / B1.17 pair names this obligation and B2.87 specifies the mechanism for discharging it.

How to cite this note

Li, W. (2026). *Evolution Mechanisms and Relational Role Determination — The Connecting Pair Where Ongoing Function Change Makes Level Redetermination a Governance Obligation (B1.12 + B1.17).* May 13, 2026. ORCID: 0009-0004-8065-3235.